

EXPLAINED EXAMPLE 5 - STEADY FLOW ENERGY EQUATION

- ① An adiabatic steam turbine extracts power from a stream of steam steadily flowing at 38.7 kg/s. At the inlet to the turbine, the steam is at a temperature of 780 K and a pressure of 10 MPa. At the exit, the fluid has a temperature of 320 K and a quality of 0.88. Calculate the power output of this turbine.

Given : $\dot{m} = 38.7 \frac{\text{kg}}{\text{s}}$

Find : $\dot{W}_2$

Assume : S.C.S.

② $T_1 = 780 \text{ K}$

$P_1 = 10 \text{ MPa}$

$T_2 = 320 \text{ K}$

$x_2 = 0.88$

①

: steady flow

: $\Delta ke = \Delta pe = 0$

: $\dot{Q}_2 = 0$

→ First law: $\dot{m}(\Delta h + \cancel{\Delta ke} + \cancel{\Delta pe}) = \cancel{\dot{Q}_2} - \dot{W}_2$

②

$\dot{m} \Delta h = -\dot{W}_2$

$\dot{W}_2 = -\dot{m} \Delta h$

→ State 1 : $T_1 = 780 \text{ K}$ } Table D.2 $P_1 = 10 \text{ MPa}$

①

$P_1 = 10 \text{ MPa}$

$T_1 > T_{\text{sat}} \rightarrow \text{vapor}$

: Table D.3P $\rightarrow h_1 = 3392.8 \frac{\text{kJ}}{\text{kg}}$

→ State 2 : $T_2 = 320 \text{ K}$

②

$x_2 = 0.88 \rightarrow \text{mixture} \rightarrow \text{Table D.1}$

$h_2 = x_2 h_g + (1-x_2) h_f$

$= (0.88)(2585.7) + (0.12)(196.17)$

$= 2298.96 \frac{\text{kJ}}{\text{kg}}$

$$\rightarrow \dot{W}_2 = -\dot{m}(h_2 - h_1)$$

①

$$= -(38.7)(2298.96 - 3392.8)$$

$$\Rightarrow \boxed{\dot{W}_2 = 42.331 \text{ MW}}$$